

Vincent Herrmann

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 [vincentherrmann](https://github.com/vincentherrmann)

Lugano, Switzerland

EDUCATION

- **PhD in Artificial Intelligence** 2020–Present
The Swiss AI Lab IDSIA, University of Lugano Lugano, Switzerland
 - Working on Artificial Curiosity, Representation Learning and Reinforcement Learning.
Supervised by Prof. Dr. Jürgen Schmidhuber
- **Master of Arts in Music Informatics** 2017–2020
University of Music Karlsruhe Karlsruhe, Germany
 - Grade: 1.2
- **Master of Music in Piano Performance** 2017–2019
University of Music and Performing Arts Stuttgart Stuttgart, Germany
 - 1.0 with distinction (best possible grade)
- **Bachelor of Music in Piano Performance and Composition** 2010–2015
University of Music and Performing Arts Stuttgart Stuttgart, Germany
 - 1.0 with distinction (best possible grade)

WORK EXPERIENCE

- **Research Consultant** 2025–Present
King Abdullah University of Science and Technology Thuwal, Saudi-Arabia
 - Supporting the research of Jürgen Schmidhuber's group at KAUST
- **Lecturer** 2022–Present
University of Music Karlsruhe Karlsruhe, Germany
 - Teaching courses on deep learning and generative modeling for music
- **Lecturer** 2021–2022
University of Music and Performing Arts Stuttgart Stuttgart, Germany
 - Teaching piano performance, substituting for Prof. Michael Hauber
- **Master Student** 2019–2020
Bosch Center for Artificial Intelligence Renningen, Germany
 - Research on generative models of symbolic music
- **Research and Teaching Assistant** 2018–2020
University of Music Karlsruhe Karlsruhe, Germany
 - Teaching tutorials on music-related AI programming
- **Research Assistant** 2016–2017
University of Music and Performing Arts Karlsruhe Stuttgart, Germany
 - Analyzing performance data from a computerized grand piano
- **Freelance Work** 2011–Present
 - Pianist, composer, arranger, and consultant for interactive live-electronic projects

ACADEMIC GUIDANCE

(Co-)supervising the Master theses at the University of Lugano of Joonsu Gha (now PhD student at ETH Zürich), Jacopo di Ventura (now PhD student at Leiden University), Suuraj Perpel (expected to graduate spring 2026), and Kun Zhan (ongoing).

PROJECTS

- **Generative Transformer-based Models of Symbolic Polyphonic Music** 2020
Master Thesis supervised by Prof. Dr. Christoph Seibert Master Thesis
◦ Grade: 1.0
- **Immersion - How Does Music Sound to Artificial Ears?** 2019
Project that explores how music is perceived by AI systems
◦ Awarded the Outstanding Demonstration Award at NeurIPS 2019

PUBLICATIONS (SELECTION)

- Vincent Herrmann, Jürgen Schmidhuber (2026). **Position: Interestingness is an Inductive Heuristic for Future Compression Progress**. Accepted for Publication at ICML 2026.
- Vincent Herrmann*, Eric Alcaide*, Michael Wandb, Jürgen Schmidhuber (2025). **Multiple Token Divergence: Measuring and Steering In-Context Computation Density**. In *ICLR 2025*.
- Vincent Herrmann, Róbert Csordás, Jürgen Schmidhuber (2025). **Measuring In-Context Computation Complexity via Hidden State Prediction**. In *ICML 2025*.
- Jacopo Di Ventura, Dylan R Ashley, Vincent Herrmann, Francesco Faccio, Jürgen Schmidhuber (2025). **Upside Down Reinforcement Learning with Policy Generators**. In *EWRL 2025*.
- Dylan R. Ashley*, Vincent Herrmann*, Zachary Friggstad, Jürgen Schmidhuber (2024). **On the Distillation of Stories for Transferring Narrative Arcs in Collections of Independent Media**. In *IEEE TPAMI*.
- Vincent Herrmann, Francesco Faccio, Jürgen Schmidhuber (2024). **Learning useful representations of recurrent neural network weight matrices**. In *ICML 2024* [Oral].
- Francesco Faccio*, Vincent Herrmann*, Aditya Ramesh, Louis Kirsch, Jürgen Schmidhuber (2023). **Goal-Conditioned Generators of Deep Policies**. In *AAAI 2023* [Oral].
- Vincent Herrmann, Louis Kirsch, Jürgen Schmidhuber (2023). **Learning one abstract bit at a time through self-invented experiments encoded as neural networks**. In *IWAI 2023* [Oral].
- Mingchen Zhuge*, Haozhe Liu*, Francesco Faccio*, Dylan R Ashley*, Róbert Csordás, Anand Gopalakrishnan, Abdullah Hamdi, Hasan Abed Al Kader Hammoud, Vincent Herrmann, Kazuki Irie, Louis Kirsch, Bing Li, Guohao Li, Shuming Liu, Jinjie Mai, Piotr Piękos, Aditya Ramesh, Imanol Schlag, Weimin Shi, Aleksandar Stanić, Wenyi Wang, Yuhui Wang, Mengmeng Xu, Deng-Ping Fan, Bernard Ghanem, Jürgen Schmidhuber (2023). **Mindstorms in natural language-based societies of mind**. In *CVM*.
- Joonsu Gha, Vincent Herrmann, Benjamin Grewe, Jürgen Schmidhuber, Anand Gopalakrishnan (2023). **Unsupervised musical object discovery from audio**. In *NeurIPS 2023 Machine Learning for Audio Workshop*.
- Vincent Herrmann (2020). **Visualizing and sonifying how an artificial ear hears music**. In *PMLR post proceedings*, NeurIPS 2019.
- Vincent Herrmann (2019). **Immersion - How Does Music Sound to Artificial Ears?**. In *NeurIPS 2019 Machine Learning for Creativity and Design Workshop*.
- Vincent Herrmann (2017). **Wasserstein GAN and the Kantorovich-Rubinstein Duality**. Blog post.
- Vincent Herrmann (2016). **Wavelets - From Filter Banks to the Dilation Equation**. Published on *dsprelated.com*.
- Vincent Herrmann (2016). **Wavelets - Vanishing Moments and Spectral Factorization**. Published on *dsprelated.com*.

* signifies shared first-authorship

HONORS AND AWARDS

- **NeurIPS R0-FoMo Workshop Best Paper Award** 2023
For "Mindstorms in natural language-based societies of mind"
- **ICML Decision-aware RL Workshop Award** 2022
For "Goal-Conditioned Generators of Deep Policies"
- **KAUST Rising Stars** 2022
Speaker at the first KAUST Rising Stars Symposium
- **NeurIPS Outstanding Demonstration Award** 2019
For "Immersion - How does Music sound to Artificial Ears?"
- **Finalist at the International Piano Competition Ferruccio Busoni, Bolzano** 2015

SKILLS

- **Machine Learning Frameworks:** PyTorch, jax
- **Machine Learning Research:** Experience with training and fine-tuning large language models, reinforcement learning agents, running large scale experiments, mechanistic interpretability and advanced visualization techniques
- **Programming Languages:** Python, C, Swift, Max/MSP, PureData, JavaScript, Java
- **Music Software:** Logic Pro, Ableton Live, Finale, Dorico
- **Other Tools:** Adobe Photoshop, Illustrator, InDesign

LANGUAGES

German: Native

English: Proficient

Italian: Basic

Latin, Ancient Greek: Rusty